

Explanation Scoring Rubric

Study: Evaluating AI Guardrails and Their Effect on Students' Independent Critical Thinking

Purpose

This training module introduces a structured approach to using AI tools effectively while maintaining independent thinking.

The goal is not to restrict AI usage, but to guide students in using AI as a **learning aid rather than a shortcut**.

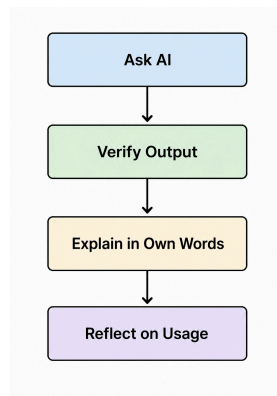
What Are AI Guardrails?

AI guardrails are simple steps that help ensure students:

- Do not blindly accept AI-generated answers
- Actively verify correctness
- Understand the reasoning behind solutions
- Reflect on their learning process

Guardrail Framework

Follow this 4-step process whenever using AI:



Step 1: Ask AI

- Use AI to understand the problem or get guidance
- Ask clear and specific questions

Example:

“Explain why this model is overfitting.”

Step 2: Verify Output

Before accepting the answer:

- Cross-check with another source
- Ask follow-up questions
- Check if the logic makes sense

Ask yourself:

- “Is this correct?”
- “Why does this work?”

Step 3: Explain in Your Own Words

- Do not copy AI responses
- Try to rewrite the explanation yourself
- Focus on understanding, not memorization

Example:

Instead of copying:

“Overfitting occurs when...”

Write:

“The model is overfitting because it memorizes training data instead of learning patterns.”

Step 4: Reflect on Usage

After solving the problem:

- Did AI help you understand?
- Could you solve it without AI next time?
- What part was confusing?

Do's and Don'ts

Do:

- Use AI as a guide
- Verify answers
- Think critically
- Ask “why” and “how”

Don't:

- Copy answers blindly
- Skip reasoning
- Depend entirely on AI
- Assume AI is always correct

Example Walkthrough

Problem:

A model performs well on training data but poorly on test data.

Without Guardrails:

- Copy AI answer
- Move on

With Guardrails:

1. Ask AI → “Why is this happening?”
2. Verify → Check another source or example

3. Explain → Write in own words
4. Reflect → “Now I understand overfitting”

Why This Matters

Using guardrails helps:

- Improve understanding
- Strengthen reasoning skills
- Reduce over-reliance on AI
- Build confidence in independent problem-solving

Summary

AI is a powerful tool, but learning happens when you:

- Question
- Verify
- Explain
- Reflect

Guardrails help turn AI into a **learning partner**, not a replacement for thinking.